

UCH302: PROCESS FLUID MECHANICS

L	T	P	Cr
3	1	2	4.5

Course Objective:

To introduce the principles of fluid mechanics and illustrate their applications in the pipe networking, fluid machinery and various flow measuring devices.

Introduction: Definition of a fluid and its properties; Classification of fluids

Hydrostatics: Measurement of pressure: Manometers; Thrust on submerged surfaces

Fluid flow phenomena: Description of fluid flow; Velocity and acceleration of a fluid particle; Continuity equation; Circulation, potential function and stream function; Energy equation: Euler and Bernoulli equations; Navier-Stokes equation; Flow measuring devices: Pitot tube, Orifice meter, Venturimeter, Rotameter; Momentum equation and its applications.

Flow through Pipe: Fully developed flow; Laminar pipe flow: Hagen-Poiseuille law; Introduction to turbulent pipe flow; Major and minor energy losses, Pipe networks: Pipes in series and parallel, Concept of equivalent pipe.

Pumps and Compressors: Types, Working principles, Basic equations, characteristic curve, Net positive suction head (NPSH), Cavitation, Priming.

Flow of Compressible Fluids: Basic equations; Adiabatic, isothermal and isentropic flows.

Dimensional analysis: Methods of dimensional analysis

Laboratory work:

Verification of Bernoulli's theorem, Calibration of Venturimeter, Calibration of orifice meter, Rheology of non-Newtonian fluids, Determination of friction factor for pipes of different materials, Determination of hydraulic coefficients of an orifice, Verification of momentum equation, Determination of loss coefficients for various types of pipe fittings, Calibration of rotameter, Visualization of laminar and turbulent flow.

Course Learning Outcomes (CLO)

Upon completion of the course, students will be able to:

1. capability to analyze and solve problems of fluid based engineering systems including pressures and forces on submerged surfaces
2. analyze fluid flow problems with the application of the mass, momentum and energy equations
3. evaluate practical problems associated with pipe flow systems
4. analyze and solve the problems related to compressible fluids, and fluid machinery.

Text Books

1. McCabe, W., Smith, J. and Harriot, P., *Unit Operations of Chemical Engineering*, McGraw-Hill (2005) 7th ed.

Reference Books

1. Fox, R.W., McDonald, A.T, and Pritchard, P.J., *Introduction to Fluid Mechanics*, John Wiley (2008) 6th ed.
2. Wilkes, J.O., *Fluid Mechanics for Chemical Engineers with Microfluidics and CFD*, Prentice Hall of India (2005) 2nd ed.
3. Denn, M., *Process Fluid Mechanics*, Prentice Hall (1979).
4. Cengel Y., Cimbala J., *Fluid Mechanics: Fundamentals and Applications*, McGraw-Hill (2005) 4th ed.

Evaluation Scheme:

S. No.	Evaluation Elements	Weightage (%)
1.	MST	25
2.	EST	45
3.	Sessional (may include Quizzes/Assignments/Tutorials/Lab evaluations)	30